

EXPERIMENT - 8

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8.1.1] Multiplication of Two Square Matrices

A) ALGORITHM

Step 1: Start

Step 2: Read the dimension n of the square matrices.

Step 3: Input the elements of first matrix A of size $n \times n$.

Step 4: Input the elements of second matrix B of size $n \times n$.

Step 5: Initialize a result matrix C of size $n \times n$ with all elements as 0.

Step 6: For i = 0 to n-1

 For j = 0 to n-1

 For k = 0 to n-1

$C[i][j] = C[i][j] + A[i][k] * B[k][j]$

Step 7: Display the resultant matrix C.

Step 8: Stop

B) PYTHON CODE

```
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    r = list(map(int, input().split()))
    A.append(r)
print("second matrix:")
B = []
for i in range(n):
    r = list(map(int, input().split()))
    B.append(r)
result = []
for i in range(n):
    result.append([0]*n)
for i in range(n):
    for j in range(n):
        for k in range(n):
```

```

        result[i][j] += A[i][k] * B[k][j]

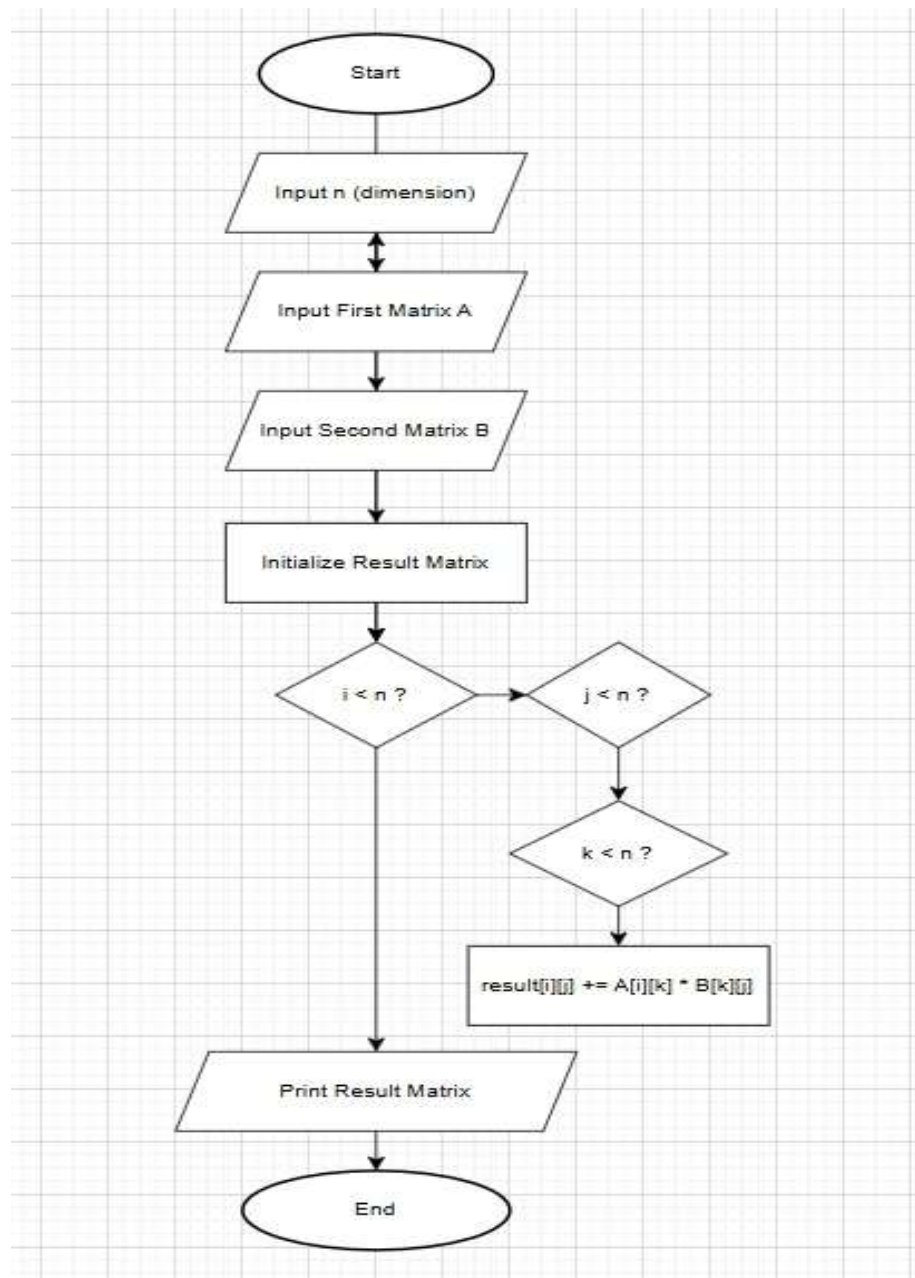
print("Resultant Matrix:")

for i in range(n):
    for j in range(n):
        if j == n - 1:
            print(result[i][j], end="")
        else:
            print(result[i][j], end=" ")

    print()

```

C) FLOWCHART



D) EXCECUTION

CODETANTRA

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8.1.1. Multiplication of Two Square Matrices

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Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n , representing the dimensions of the square matrices ($n \times n$).
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases

Multiplication...

Debugger

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1 # Type Content here...

2 n = int(input("dimension: "))

3 print("first matrix:")

4 A = []

5 for i in range(n):

6 r = list(map(int, input().split()))

7 A.append(r)

8

9 print("second matrix:")

10 B = []

11 for i in range(n):

Average time

0.029 s

Maximum time

0.043 s

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 25 ms

Debug

Expected output

Actual output

dimension: 2

first matrix:

1 2

3 4

second matrix:

5 6

dimension: 2

first matrix:

1 2

3 4

second matrix:

5 6

Terminal

Test cases

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